Design notes: steel mitre gates for Canal Heritage

Chosen design. Each end of each lock gets a pair of steel mitre gates (design A). The two leaves meet at a 1 in 3 mitre pointing upstream. A vertical lift gate (design B) was also drawn, but its 11.8 m towers would dominate a heritage canal and its winch could not be hidden in a chamber, so it was rejected.

Size and water. The gates fit the 4.2 m opening, are 6.0 m high and are designed for the worst case of 5.5 m of water on one side and none on the other. That puts 54.0 kPa on the cill and 365 kN on each leaf. Water pushes the leaves together at the mitre and back into the quoins, so the gate works as an arch and gets tighter as the water rises. Eight girders, closer together at the bottom where the pressure is highest, carry the load. The worst skin plate panel reaches 34 MPa, a safety factor of about 10.

Automatic and independent. Each leaf has its own electric actuator, controller, position sensor and water level sensors, so the gates can be opened and shut one at a time from the off-site control centre. An interlock stops a gate moving unless the levels either side are equal.

Drive out of sight. The actuators sit in purpose-built precast concrete inspection chambers beside each gate, under a flush grating cover. Electric drives use the mains supply the brief offers and carry no hydraulic oil that could leak into the canal.

Surveillance and emergency stop. A CCTV mast on each chamber watches the gate and the lock. Leaf position, motor current and water levels are sent to the control centre. An emergency call point on each side stops both gates and opens an intercom to the lock keeper.

Boat strikes. Replaceable oak fenders on the chamber face take any knocks, the gates only move when the CCTV shows the lock clear, and the cill is marked so boats stay clear of it.

Inspection. The walkway on each leaf, and the chamber covers, give safe access to look at the hinges, actuators and seals.

Hinges. The leaf turns on a 150 mm stainless pintle in a bronze bush at the bottom and a journal in an anchor collar at the top. Stainless and bronze do not corrode together, and the FEA gives a safety factor of 12.

Seals. EPDM rubber seals run along the cill, behind the heel post and between the mitre posts. Water pressure squeezes them shut. They bolt on, so they can be changed without welding.

Removal and maintenance. Each leaf weighs 8.7 t and is lifted out by mobile crane once the anchor collar is released. There is nothing to grease daily: routine work is a yearly look and a check of the seals and bush.

Looks and sustainability. The gates keep the traditional mitre shape, painted black and white with oak fenders. Steel, stainless steel, bronze and oak can all be recycled or reused, and the oak comes from FSC certified forests.

Budget and time. Estimated supply price about £95,000 per gate system (steel fabrication, actuators, hinges, seals, controls and chambers), inside the £100,000 budget. Ten systems can be designed, made and fitted within 52 weeks.